JSynth

The flocking synth. User Manual



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0.15.0	CLAP · Instrument · MPE	macOS · Windows · Linux	October 2026

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About this manual

Chapters 2 and 3 get you installed and playing in a few minutes. Chapter 4 tours the panel, and Chapter 5 explains how a note becomes a sound, which is the quickest way to understand what every control does. Chapters 6 to 13 cover the sky, the parts and every panel. Chapter 14 lists the factory presets and Chapter

15 has recipes to start from. The appendix lists every parameter with its exact range and default, plus specifications and troubleshooting.

Throughout the manual, control names are set in **bold** and panel labels as they appear on screen, such as **TUNE PULL**. Every control is a host parameter, and every one that moves continuously can be modulated by the host. A tag beside a parameter means:

ROUTE A route can move it, per note, and the host can modulate it per note too.

CHAPTER 01

Welcome to JSynth

A two-part MPE synth where every note is a creature, and the creatures flock.

Play a note on JSynth and a creature hatches in the sky in the middle of the panel. Your note is its **lure**: low notes pull their creatures to the left, high notes to the right, and sliding your finger up an MPE controller lifts it higher. Press harder and the creature flies faster and burns brighter. Let go and it flies off as its sound fades, trailing ink.

While they are held, the creatures do what flocks do. They gather, turn together, keep a little apart, and wander. And because each creature is a voice, how it flies can become how it sounds: how fast it goes, how near its nearest neighbour is, which way it heads and how hard it turns are all modulation sources, as exact as an envelope. Creatures can also hear each other directly. A neighbour's oscillator can frequency-modulate this one, creatures flying the same way drift toward pure intervals, and a hunter that catches its prey rings it.

Underneath is a serious synth. Each of the two parts, **A** and **B**, is its own species with its own patch: three oscillators with analog and FM waves, a 3×3 FM grid, noise, drive, two filters, three envelopes, four LFOs, four mod buses and sixty-four routes. Nothing the flock does reaches the sound unless you route it, so a plain patch is plain.

At a glance

- **Two parts**, layered, split, by velocity, by MPE zone, or crossfaded by expression. Up to 16 voices each.
- **Three oscillators** per voice: sine, triangle, saw, square, pulse, the TX81Z's FM waves and noise, pitched by note, ratio or fixed Hz.
- **A 3×3 FM grid**, phase or exponential, whose diagonal is each oscillator's feedback.
- **Two filters** per voice: SVF up to 48 dB, notch, peak, ladder and comb, in serial or parallel, with each oscillator picking its filter.
- **MPE** and CLAP note expressions: velocity, press, glide, slide and release velocity, each with its own curve and smoothing.
- **The flock**: boids, each creature following its neighbours by cohesion, alignment and separation, plus speed, wander, lure and range, with four behaviours: Shoal, Scatter, Hunt and Flee.
- **Four flock sources** per creature: Speed, Nearest, Heading and Turn.
- **Between creatures**: Neighbour FM, Tune Pull and Bite.
- **Shape envelopes** after XAOC's Zadar, 60 contours from 1 ms to 30 minutes, and **four mod buses** per part.
- **MAP**: touch a source, drag a knob. No matrix page, and up to 64 routes per part.
- **116 factory presets** in 12 categories: keys, organs, mallets, plucks, basses, leads, pads, strings and brass, sequences, drums and effects that play straight from a keyboard, then 22 the flock plays itself, each in its own room.

> **THE FLOCK IS THE ENGINE'S**
> The flock flies on the audio thread, at a fixed rate, inside the synth. What you see in the sky is exactly what the voices hear, and a render plays back the same every time from the same start.

CHAPTER 02

Installation

JSynth is a CLAP plug-in. It ships in the JClap bundle alongside the other JClap plug-ins.

Requirements

Plug-in format	CLAP 1.2 (instrument, stereo). Your host must support CLAP, for example Bitwig Studio, REAPER, FL Studio or Studio One.
macOS	Universal binary: macOS 10.15 or later on Intel, 11 or later on Apple Silicon.
Windows	64-bit Windows.
Linux	64-bit, X11.
Controller	Any MIDI keyboard. An MPE controller (a Seaboard, Linnstrument, Osmose or similar) or a host that sends CLAP note expressions brings out the most.

Installing

01 **Download**

Grab the zip for your operating system from the latest release at `github.com/jeffmhopkins/Jclap/releases/latest`. It contains every JClap plug-in.

02 **Copy the plug-in**

Copy `JSynth.clap` (a bundle on macOS) into your CLAP folder:

macOS	`~/Library/Audio/Plug-Ins/CLAP`
Windows	`C:\Program Files\Common Files\CLAP`
Linux	`~/.clap`

03 **Rescan**

Restart your host or rescan plug-ins. In Bitwig Studio this is under *Settings → Locations*. JSynth appears as an instrument.

MACOS: BROWSER DOWNLOADS ARE QUARANTINED

The plug-ins are ad-hoc signed but not notarized, so macOS quarantines files downloaded through a web browser and some hosts (Bitwig among them) silently skip them. The simplest install is from Terminal, since files fetched with `curl` are not flagged:

```
curl -sL https://github.com/jeffmhopkins/Jclap/releases/latest/download/jclap-macOS.zip -o /tmp/jclap.zip
&& mkdir -p ~/Library/Audio/Plug-Ins/CLAP && ditto -x -k /tmp/jclap.zip ~/Library/Audio/Plug-Ins/CLAP
```

If you already downloaded with a browser, clear the flag once after copying:

```
xattr -dr com.apple.quarantine ~/Library/Audio/Plug-Ins/CLAP/*.clap
```

Outputs

JSynth has one note input and three stereo outputs: **Main**, **Part A** and **Part B**. Normally both parts play on Main. Set a part's **Output** to *Own* (a host parameter) and it plays on its own port instead, so you can process the two species separately in your host.

CHAPTER 03

Quick Start

Five minutes from an empty instrument slot to a flock you can hear.

01 **Load JSynth and play**

Put JSynth on an instrument track and play a few notes. Each note hatches a creature at its lure: low notes on the left, high on the right. Hold a chord and watch them gather; let go and they fly off as they fade. No keyboard to hand? Click in the sky: left to right picks the note (C2 to C6), and dragging bends it.

02 **Pick a preset**

Click the preset name in the header to open the menu, grouped by category: *Keys*, *Organs*, *Bells and Mallets*, *Plucks*, *Bass*, *Leads*, *Pads*, *Strings and Brass*, *Arps and Sequences*, *Percussion* and *FX*, then *Flock*, the sounds the creatures play themselves. Or click the arrows, or scroll over the bar, to step through them. Play a few chords on *Tine Piano* and *Warm Analog Pad*, a line on *Saw Lead* with the mod wheel up, then hold a chord on *Slow Murmuration* and *Predator Weather*.

03 **Let the creatures hear each other**

Load *Init* and turn **OSC 2**'s **LEVEL** up to about half, since Neighbour FM plays through Osc 2. Then open the **FM** tab at the top left and turn up **FLOCK FM**. Now hold a chord: when creatures fly close together they frequency-modulate each other, and the sound rings and shifts as the flock moves. Raise **TUNE PULL** too and creatures flying the same way drift into just intonation.

04 **Map a flock source**

Click **SPEED** under the sky, which turns **MAP** on with Speed armed. Then drag Filter 1's **CUTOFF** up on the **FILTER** panel at the left. A ring in Speed's colour shows the route while Speed is armed. Click **SPEED** again to finish. Now the faster a creature flies, the brighter it sounds; play harder (Press makes creatures faster) to hear it.

05 **Add a second species**

Set the play mode in the header to **LAYER** and click **PART B**. Part B is a whole second patch, and by default it hunts: its creatures chase part A's. On part B's **FM** tab, raise **BITE**, and every creature B catches rings with its hunter's tone.

06 **Save it**

Click **SAVE** in the preset bar, type a name and press Enter. Your preset appears under *User* in the preset menu. Settings are also saved with your project automatically.

> **TIP**
> Every control has a tooltip. Hover over anything on the panel for a one-line reminder of what it does.

CHAPTER 04

The Panel

The sound sits down the left in three big panels that draw what they do, the sky fills the right, and the voice and the output sit under it. Everything edits the part you have chosen.



1. **Preset bar.** Step with the arrows, click the name for the menu, **SAVE** your own.
2. **PART A / PART B.** Which species the panels edit.
3. **Play mode.** Which part each note plays.
4. **MAP.** Mapping: touch a source, drag a knob. Touching a source turns it on too.
5. **Touch chips.** VELOCITY, PRESS, GLIDE, SLIDE, RELEASE: sources that light as you play.
6. **OSCILLATORS / FM.** The three oscillators with their waves drawn; the FM grid, the links between creatures and the noise.
7. **FILTER.** Each filter's response over its own knobs and the two together, drawn live, and their controls.
8. **ENVELOPES / LFOS / BUSES.** ENV 1 AMP, ENV 2 and ENV 3; LFO 1 and 2; the four mod buses.
9. **The sky.** Both species, their ink, and the links between them. Click to play. Over its top left, the look (SUMI or AURORA) and the palette.
10. **Flock sources.** The creatures' own modulation sources, and Random.

11 **RINGS, ROUTES.** Lure rings and the route list.

12 **VOICE / FLOCK / MOTION / TOUCH.** The voice, how the species flies, the wingbeat and the wind, and how each touch responds.

13 **Meters.** The levels of A, B and the output as they play, with each one's setting marked.

14 **SYNTH / SKY.** The two views: the panels beside the sky, or the sky alone.

Working with the controls

Every knob, switch and menu is a host parameter, so everything can be automated. Knobs work the same everywhere on the panel:

ACTION	ON A KNOB
Drag up / down	Change the value. A popup shows the value while you drag.
`Shift` **+ drag**	Fine adjustment.
Scroll wheel	Step the value up or down.
Double-click	Reset to the default value.
Right-click	Menu with *Reset to default* and *Enter value…* to type an exact number.

In MAP mode, with a source armed, each knob that source moves draws a **route ring** in its colour outside its own arc, spanning how far the source moves it. Outside MAP the knobs show no rings; the route list shows every route. In MAP mode, knobs behave differently: see *Chapter 13*.

Colours

Part A is drawn in ● orange and part B in ● blue, in the sky and on the panels. Every modulation source has its own colour, the same on its chip, its route rings and in the route list, and MAP and the route list are ● pink.

Panel size

The panel is 1648 × 940 at 100%. Right-click any empty part of the sky and choose *UI Size* to scale it to 75%, 100%, 125%, 150% or 200%; at 75% it fits a 1440 × 900 laptop screen. The size you pick is remembered the next time the editor opens.

CHAPTER 05

How JSynth Works

A note becomes a voice and a creature at once. The voice makes the sound; the creature flies, and how it flies can shape the sound.

One voice

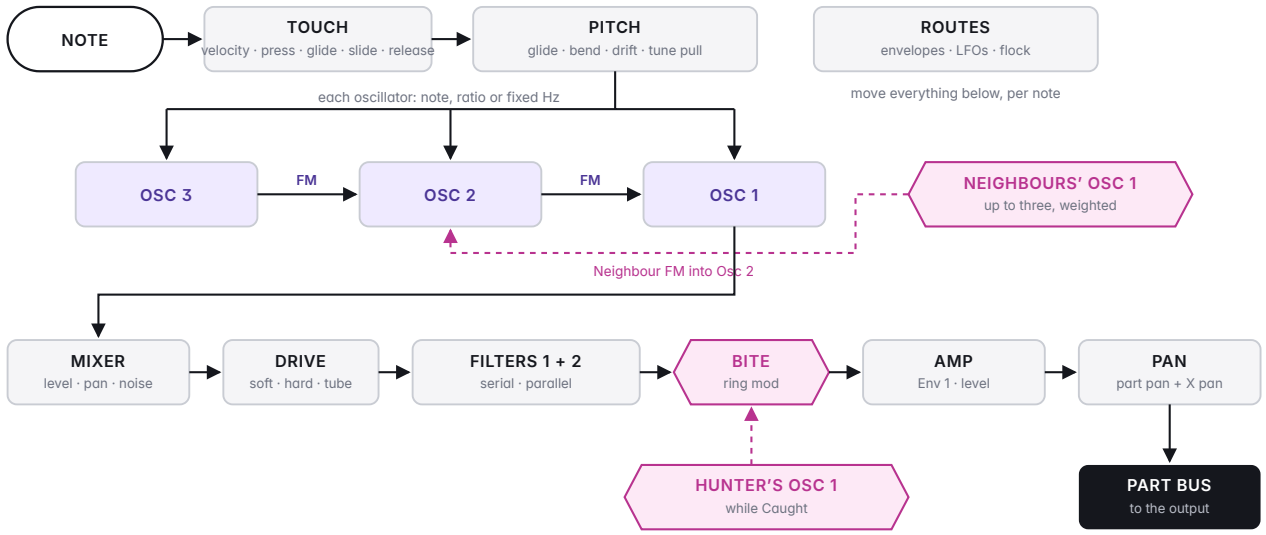


Fig. 5.1 One voice, at the oversampled rate. The pink paths are the couplings to other creatures: a neighbour's Osc 1 into Osc 2, and a hunter's Osc 1 ringing the voice.

Each voice runs three **oscillators**, in the order 3, 2, 1, so Osc 3 can modulate Osc 2, and both can modulate Osc 1, within the same sample. The **FM grid** sets how much each one modulates each other one. The **mixer** sets each oscillator's level and pan and adds noise. Then comes **drive**, then two **filters**, then the **amp**, whose shape is Env 1, and finally the voice's place in the stereo field. Every voice of a part sums to that part's bus, and the two parts sum to the output.

The control rate

Envelopes 2 and 3, the LFOs, the touch, the routes and the flock all work at a control rate of one tick every 32 samples (1,500 a second at 48 kHz). Every value the audio reads ramps smoothly across each tick, so nothing zippers, and knobs are smoothed over 10 ms. The amp envelope runs every sample. Because the ticks are counted across the host's blocks, JSynth sounds the same at any buffer size.

The creature

As a voice starts, its creature hatches at the note's lure. The flock steps 500 times a second. Each step, every creature feels its neighbours (cohesion, alignment, separation), its lure, its own wander, the wind and its wingbeat, and moves. Then the flock works out each creature's four sources and who can hear whom. Each voice reads its creature's sources smoothly between steps.

Routes

A **route** connects a source to a destination with an amount. Each part has 64 of them. Each tick, every voice works out its routes from its own sources, so the same route can push one note's filter open and leave another's closed: an envelope, a finger's pressure or a creature's speed is always that note's own. A route moves a parameter within its own range, so +100% from a source at full takes any knob from its lowest to its highest.

Quality

The voices run at 2× the host's rate by default, which keeps FM and hard waves clean, and are brought back down by a steep, latency-free filter. The **Quality** parameter (a host setting) chooses 1×, 2× or 4×. 1× saves CPU; 4× is for bright FM at low sample rates.

> **WHERE TO READ MORE**
> `plugins/jsynth/ARCHITECTURE.md` in the source repository walks through every stage with the numbers the code uses.

CHAPTER 06

The Sky

Both species fly here, all the time. The sky is the synth's own flock, drawn as it is.



Fig. 6.1 Part A's creatures and part B's in a layered patch, with their ink, in the Sumi look. Ripples are lures; a pink ring marks a creature a hunter has caught; dotted lines join neighbours that hear each other.

What you see

Creatures In Sumi a brush stroke, thick at the head and tapering behind, its width breathing with its Wingbeat LFO; in Aurora a four-pointed star trailing a thin line, its points twinkling with the wingbeat. It is brighter the harder you press, and fades with its sound after you let go. The part you are editing flies in front; the other is dimmed.

Ink Every creature sheds ink behind it, which fades by the part's **TAIL** and spreads by its **SPREAD**, taking on a deeper colour as it ages. In Sumi it marbles into fine filaments; in Aurora each wisp hangs a curtain of light above it. Creatures flying through ink stir it, and so does the mouse.

Lures Where each held note pulls its creature: ripples in Sumi, wider the harder you press, and a small diamond over a faint plumb line in Aurora.

Links	From a creature to the neighbour it hears most, brighter the more **FLOCK FM** it gets: dotted in Sumi, an arc with a pulse running along it in Aurora. A hunter has a line to what it has caught.
Readout	Hover a creature to read which voice it is and its Speed, Turn and Nearest.
Caught	A pink ring around a creature a hunter holds.

Looks and palettes

The switch over the sky's top left picks how it is drawn. **SUMI** is ink in dark water, lit from above; **AURORA** is a night sky over a glowing horizon. The menu beside it picks a palette: **EMBER**, **BOREALIS**, **GLACIER**, **SUNSET**, **MOONLIGHT**, **NEON** or **CLASSIC**, each with a set of inks for each part. Each look keeps its own palette (Sumi starts in Ember, Aurora in Borealis), and the look and both palettes are saved with your project. **SIZE** beside the palette draws the creatures, their strokes and their ink bigger or smaller, from 50% to 300%, and is saved with your project too. They are only drawn and never change the sound.

Playing the sky

Click anywhere in the sky to play a note: left to right runs from C2 to C6, and how high you click sets Slide. Drag while holding to bend the note and change Slide; let go to release it. The notes play on whichever parts the play mode sends them to, just like notes from your keyboard.

The flock sources

The row of chips under the sky holds each creature's four sources, and **RANDOM**, a new value for each note. Each chip's light shows the newest creature of the part you are editing. They are always each note's own when used in a route.

SOURCE	WHAT IT MEASURES
SPEED	How fast the creature flies, against the fastest it could at full Press (0 to 100%).
NEAREST	How near its nearest neighbour is: 100% is touching, 0 is at the edge of **RANGE** or beyond.
HEADING	How much it flies upward: 100% straight up, 50% level, 0 straight down. Smooth all the way round, so it never jumps.
TURN	Which way and how hard it is turning: up to +100% for a hard right turn on screen, down to −100% for a hard left, 0 flying straight. Smoothed over about 170 ms, so the wander's jitter and the wingbeat's bob barely reach it.

Two views

The switch at the top right picks how much of the panel you see. **SYNTH**, as this manual shows it, puts the sound down the left beside the sky. **SKY** folds everything away but the switch, the look and the preset bar so the sky fills the panel, which is lovely on a second screen while you play. There those controls fade out after 10 seconds without the mouse moving over the window, and come back as soon as it moves. The view stays as you left it when the editor is closed and opened again.

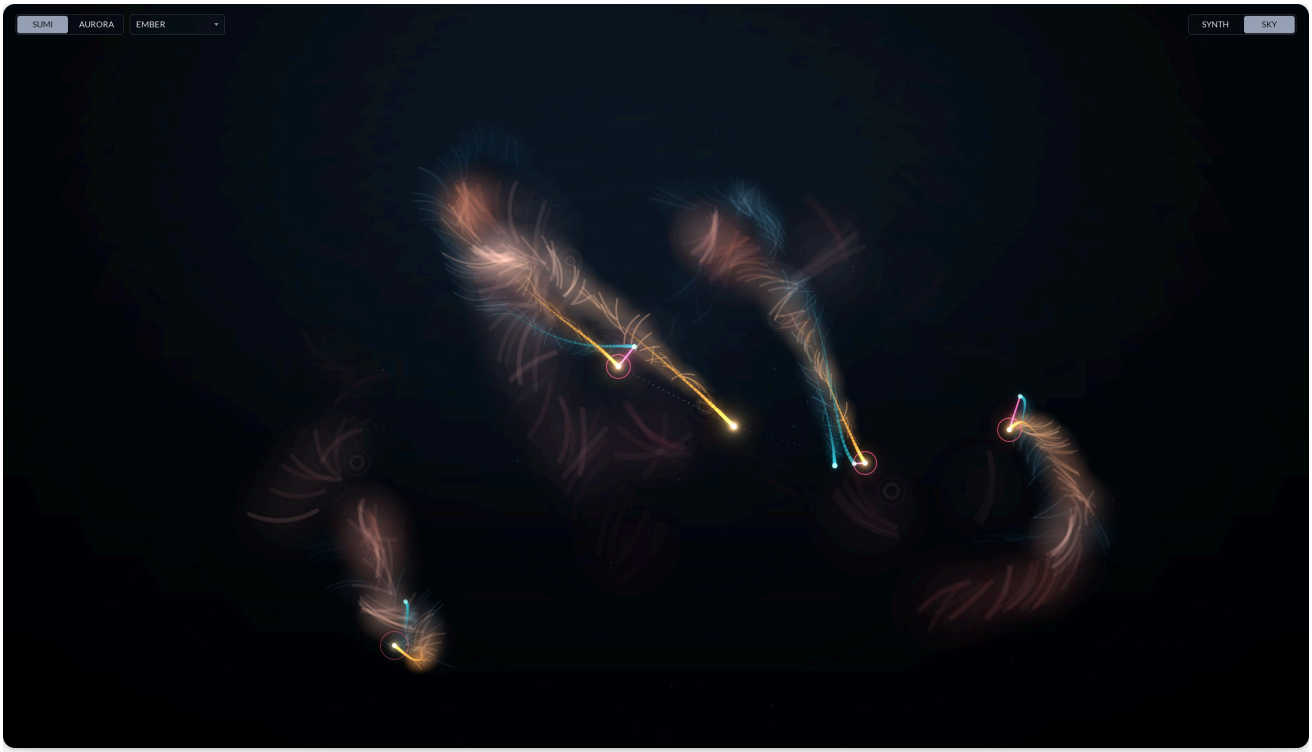


Fig. 6.2 SKY: the flocks alone.

CHAPTER 07

Parts and Playing

Two species, one keyboard. The header decides which notes go where; the VOICE tab under the sky sets how each part plays them.



Fig. 7.1 The header: logo, preset bar, PART A / PART B, play mode, MAP, the touch chips and the view switch.

Two parts

Parts A and B are complete, separate patches with the same controls. **PART A** and **PART B** in the header choose which one the panels show and edit; the sky always shows both. A new instance starts with A shoaling and B hunting it an octave down, so layering them gives you a predator from the start.

Play modes

MODE	WHICH PART PLAYS A NOTE
SINGLE	Part A only. The default.
LAYER	Both, every note.
SPLIT	B below the **Split Key** (C4 by default), A from it up.
MPE ZONES	MPE member channels 2 to 8 play A and 9 to 16 play B. Channel 1 plays both.
VELOCITY	B for notes struck at or above the **Velocity Split** (80), A for softer ones.
EXPRESSION	Both, crossfaded per note by Slide or Press (the **Crossfade Source**): at 0 you hear A, at full you hear B.

Split Key, Velocity Split and Crossfade Source, and each part's Key Low and Key High range, are host parameters. A note outside a part's key range never plays that part.

VOICE



The VOICE tab draws what its knobs set: the stereo field (the part's pan, its X Pan spread, the output's width and where its creatures sit), the voices in use, and at the right the **meters**. These show the levels of A, B and the whole output as they play, from −60 to +6 dB, glowing brighter the louder they are and red above 0 dB. A white tick on each marks where its LEVEL or MASTER is set. The part you are editing is lit; the other is dimmed.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
POLY / MONO / LEGATO	`Voice mode`	`Poly`	Mono plays one note at a time and restarts it; Legato glides to a new note without restarting while one is held. Both go back to the last note still held when you let go.
Voices (– 8 +)	`1 – 16`	`8`	Polyphony: how many notes the part sounds at most. The oldest released note is taken first, then the oldest held, with a 3 ms fade.
PAN	`±100%`	`0`	The part's pan. ROUTE
X PAN	`0 – 100%`	`0%`	How far each creature's place in the sky, left to right, pans its voice. At 100% a creature at the left edge plays hard left. ROUTE
WIDTH	`0 – 200%`	`100%`	The whole output's stereo width: 0% is mono, past 100% wider. Shared by both parts.
GLIDE	`0 ms – 20 s`	`0`	Portamento time. ROUTE
BEND	`1 – 96 st`	`48 st`	The MPE bend range: how far a finger slides on a member channel.
TRANSP	`±48 st`	`0 (B: -12)`	Transpose, in semitones.
VELOCITY	`0 – 100%`	`100%`	How much velocity sets the level. At 0 every note plays at full level.
LEVEL	`-60 – +6 dB`	`0 dB`	The part's level. ROUTE

CONTROL	RANGE	DEFAULT	WHAT IT DOES
MASTER	`-60 – +6 dB`	`-6 dB`	The whole output's level. Shared by both parts.

Host parameters that complete the part: **Fine** (±100 cents), **Glide Mode** (Always, or Legato: glide only from a note still held), **Key Low** and **Key High**, and **Output** (Main or Own).

MPE and MIDI

JSynth takes CLAP notes, plain MIDI and MIDI MPE.

- **Channel 1** is the MPE manager channel and a plain keyboard's channel. Its pitch bend moves every note by the **MGR BEND** range (2 semitones by default, on the TOUCH tab). CC 1 is the Mod Wheel source and CC 64 is sustain. If it sends channel pressure or CC 74, those reach every note too.
- **Channels 2 to 16** are member channels, each carrying one note's own bend (by the part's **BEND** range), pressure and CC 74.
- **CLAP notes** use note expressions instead: tuning, pressure and brightness.

Hosts that support CLAP per-note modulation can also modulate any parameter tagged ROUTE on a single note.

CHAPTER 08

Oscillators and FM

Three oscillators per voice, from soft sines to band-limited saws, and a grid that lets each one modulate the others.



Fig. 8.1 The OSCILLATORS panel: a row for each oscillator, with its wave drawn as its Wave and Shape make it.

An oscillator

CONTROL	RANGE	DEFAULT	WHAT IT DOES
NOTE / RATIO / HZ	Pitch mode	Note (Osc 2, 3: Ratio)	**Note** follows the keyboard with an offset; **Ratio** runs at a multiple of the note, the classic FM way; **Hz** holds a fixed frequency whatever you play.
Wave	15 waves	Saw (Osc 2, 3: Sine)	See the table below.
RAND / SYNC	Retrigger	Sync	**SYNC** starts each note's wave at its **Phase**; **RAND** at a random place, so no two notes start alike.
PITCH	±48 st	0	Note mode: the offset in semitones. ROUTE
RATIO	×0.25 – ×16	×1 (Osc 3: ×3)	Ratio mode (the knob reads RATIO): the multiple of the note. Routes move it by musical intervals. ROUTE
FREQ	1 Hz – 10 kHz	440 Hz	Hz mode (the knob reads FREQ): the fixed frequency. ROUTE
FINE	±100 ct	0	Fine tune, in every mode. ROUTE
SHAPE	0 – 100%	50%	Pulse width for Square and Pulse; for the other waves (except Saw), a phase distortion that squeezes or stretches the first half of the cycle. 50% is the plain wave. ROUTE

CONTROL	RANGE	DEFAULT	WHAT IT DOES
LEVEL	`0 – 100%`	`100% (Osc 2, 3: 0%)`	How loud it is in the mix. An oscillator at 0 can still modulate others. ROUTE
PAN	`±100%`	`0`	Its place in the voice's stereo image. ROUTE
FILTER	`1+2, F1, F2`	`1+2`	Which filter it goes into. See *Routing* in Chapter 09.

Host parameters per oscillator: **Sync** (Osc 2 and 3 restart with Osc 1), **Phase** ROUTE, **Bend** (how much the pitch bend moves it, 100% by default) and **Key Track** (Off pitches it from middle C whatever you play).

The waves

WAVE	CHARACTER
Sine	Pure. The classic FM operator.
Triangle	Soft, odd harmonics.
Saw	Bright and full. Band-limited, so it stays clean high up and under FM.
Square, Pulse	Hollow to nasal. Shape sets the width: 10 to 90% for Square, 5 to 50% for Pulse.
Half Sine … Alt Camel	The Yamaha TX81Z's seven other waves: Half Sine, Abs Sine, Quarter, Alt Sine, Alt Half, Camel and Alt Camel. Sines with parts of the cycle cut, folded or doubled, which give FM a whole family of brighter, buzzier tones.
White, Pink	Noise, as an oscillator: it can be filtered, panned and even used as an FM source.
Random	A different one of Sine to Alt Camel for each note, so a run of notes changes colour as it goes.

OSCILLATORS

FM



The **FM** tab draws what comes out of Osc 1, the oscillator you hear on its own, for a C4, with a chip for each FM cell that is turned up. The grid reads **row into column**: the knob in row 2, column 1 (**2 > 1**) is how much Osc 2 modulates Osc 1. The diagonal, **FB 1** to **FB 3**, is each oscillator's feedback on itself. Each knob is squared inside, so the first half of its travel is gentle.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
FB, 1 > 2 … 3 > 2	0 – 100%	0%	The nine cells. ROUTE
PHASE / EXP	FM type	Phase	**Phase** is DX-style phase modulation, bright and stable in pitch. **Exp** moves the pitch itself, up to four octaves either way: growly, and it bends.
FLOCK FM	0 – 100%	0%	Neighbours' Osc 1 into this creature's Osc 2. ROUTE
TUNE PULL	0 – 100%	0%	Creatures flying the same way drift toward pure intervals. ROUTE
BITE	0 – 100%	0%	Ring mod while this part's hunters hold their prey. ROUTE
NOISE	0 – 100%	0%	The mixer's noise level. ROUTE
COLOR	±100%	0	Pink (−100%) to white (+100%). ROUTE
DRIFT	0 – 100%	0%	Each creature's own slow wander in pitch, up to ±15 cents. ROUTE

Order and feedback

Within each sample the oscillators run 3, then 2, then 1. So the downward cells (3 > 2, 3 > 1 and 2 > 1) hear this sample's modulator, and the upward ones (1 > 2, 1 > 3, 2 > 3) hear the last sample's. In practice this means you can build any of the classic stacks: 3 > 2 > 1 as a three-operator stack heard on Osc 1, or 2 > 1 and 3 > 1 as two modulators on one carrier. Feedback averages its last two samples, as the DX7 did, so it brightens toward a saw and then a rasp rather than straight into noise.

BETWEEN CREATURES (Flock FM, Tune Pull and Bite) is explained in *Chapter 11*.

CHAPTER 09

Filters and Drive

Two filters per voice, from a clean 48 dB low pass to a ringing comb, after a drive stage.

LEFT

FILTER



The graph is split to line up with the knobs under it: Filter 1's response over Filter 1's knobs, Filter 2's over Filter 2's, each with its own handle, and **OUT**, the two together as the voice hears them, over Balance and Pre Drive. Drag a numbered handle across to set its cutoff and up or down for its resonance; the wheel sets its drive and a double-click resets it. In MAP mode, with a source armed, the drag sets that source's route to the cutoff instead.

While a note plays, the curves follow what the newest note's filters are really doing: its envelopes, LFOs, routes and key tracking all move them, and a faint line stays where the knobs alone would put them. The handles stay on the knobs' settings.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
SERIAL / PARALLEL	`Routing`	`Serial`	See below.
NOISE INTO	`1+2, F1, F2`	`1+2`	Which filter the noise goes into.
Type	`8 types`	`Low Pass (F2: Ladder)`	See the table below.
Slope	`12 – 48 dB`	`24 dB`	Low, high and band pass only.
CUTOFF	`20 Hz – 20 kHz`	`1.26 kHz (F2: 20 kHz)`	For a comb, its tuning. ROUTE

CONTROL	RANGE	DEFAULT	WHAT IT DOES
RES	0 – 100%	20% (F2: 0%)	Resonance; for a peak, its boost; for a comb, its feedback. ROUTE
DRIVE	0 – 100%	0%	Saturates the filter's own input. ROUTE
KEY	0 – 100%	0%	Key tracking: at 100% the cutoff moves an octave per octave, around middle C. ROUTE
Drive type	Soft, Hard, Tube	Soft	The drive before both filters.
PRE DRIVE	0 – 100%	0%	Up to 24 dB into the shape, with the level brought back down. ROUTE
BALANCE	±100%	0	In Parallel, turns Filter 1 or Filter 2 down. ROUTE

Filter types

TYPE	SOUND
Low Pass	The workhorse: a state variable filter, cascaded for 12, 24, 36 or 48 dB per octave, with a resonance peak up to a whistle.
High Pass	The same, from the other side: thins and brightens.
Band Pass	Only a band around the cutoff. Narrower and more vocal with resonance.
Notch	Cuts a band out. Resonance narrows it. Slope is ignored.
Peak	Boosts a band. Resonance is the boost, up to five times.
Ladder	A four-pole transistor ladder: warm, thick, and it squelches and self-oscillates near full resonance.
Comb +	A short tuned delay with positive feedback: rings at every harmonic of the cutoff, like a plucked string or a tube.
Comb –	Negative feedback: rings at the odd harmonics, hollower, an octave lower.

Routing

Serial Filter 1 into Filter 2. Filter 2 starts as an open ladder, so it changes nothing until you turn it down, and costs nothing while it stays open.

Parallel Each filter hears what goes into it, and **BALANCE** turns one side down.

Each oscillator's **FILTER**, and the noise's, picks where it goes: **F1**, **F2** or both (**1+2**, the default). In Serial, F1 and 1+2 go through both filters and F2 skips Filter 1. In Parallel, a source in one filter plays at full level with BALANCE at centre, and turning BALANCE the other way turns it down. A source in both crossfades: half into each at centre, all into one at either end.

> **TWO SOUNDS IN ONE**
> Set Osc 1 to **F1** and Osc 2 and 3 to **F2**, in Parallel, to filter a body and a bright layer apart: a low pass on one, a comb or high pass on the other.

Drive

The drive sits before the filters. **Soft** is a tanh curve, round and warm. **Hard** clips flat, buzzier. **Tube** is asymmetric, so it adds even harmonics. Its gain is compensated, so turning it up thickens and compresses the sound more than it makes it louder.

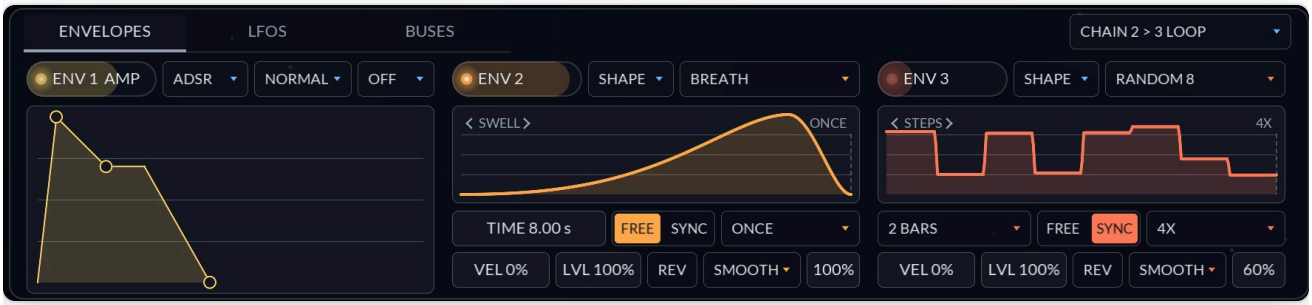
CHAPTER 10

Envelopes, LFOs and Buses

Three envelopes and four LFOs per voice, and four mod buses per part. Two of the LFOs move the creatures as well as the sound.

BOTTOM LEFT

ENVELOPES



Each envelope is an **ADSR** or a **SHAPE**, picked in the box beside its chip. As an ADSR it draws its stages: drag the points to set **attack**, **decay**, **sustain** and **release**. A point moves with the mouse from where you grab it; hold **Shift** for fine moves. A dot rides the curve for every sounding note. The chip above each one is its source for MAP and lights as it moves.

ENV 1 AMP is the amp: it shapes every note's level, and it always releases when you let go. **ENV 2** and **ENV 3** are free to route; a new part starts with Env 2 opening Filter 1.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
Attack	`0 ms – 20 s`	`5 ms`	ROUTE
Decay	`0 ms – 20 s`	`400 ms (2, 3: 300 ms)`	ROUTE
Sustain	`0 – 100%`	`70% (2, 3: 30%)`	ROUTE
Release	`0 ms – 20 s`	`300 ms`	ROUTE
Time Range	`Short, Normal, Long`	`Normal`	How much time the drawing spreads across its width: up to 1 s, 5 s or 20 s for each of attack, decay and release. Pick **Short** for plucks and drums, **Long** for pads and swells. It only changes the view: switching it never changes a time, and a time past the end waits there until you drag it back in.
Loop	`Off, Held, Free`	`Off`	**Held** repeats attack and decay while the note is held. **Free** keeps repeating after you let go too, until the note ends.

VELOCITY on the VOICE tab under the sky (0 to 100%, 100% by default) is how much velocity sets ENV 1's level. At 0 every note plays at full level.

Host parameters per envelope: **Delay** and **Hold** (0 ms to 20 s) ROUTE, an **Attack**, **Decay** and **Release Curve** each (−1 to +1), and **Velocity**, how much velocity scales the whole envelope.

Shape mode

After XAOC's Zadar module, a **SHAPE** envelope plays one drawn contour instead of its stages: pick it from **60 shapes** in ten banks, *Rise*, *Fall*, *Swell*, *Pulse*, *Waves*, *Steps*, *Drift*, *Strike*, *ADSR* and *Acoustic*, and it is stretched over its time. Click the shape box for a menu by bank, scroll over it, or use the arrows either side of the bank's name on the drawing to step through them (the name itself opens the menu). Long, slow shapes on a filter or a drawbar keep a held chord moving for minutes; a pulse or a step shape as ENV 1 turns each held note into a pattern of its own.

The drawing shows the shape as it plays, with its bank and how often it repeats in its corners and a dot for every sounding note. **Drag it sideways to warp it** in time and **up or down to bend it** in level (Shift for fine moves; double-click puts both back). In MAP mode, with a source armed, the same drag sets how far that source warps it (sideways) or bends it (up or down), and the shape as the source at its fullest would take it shows over it in the source's colour; right-click removes both routes.

Sustain point. The dashed line on the drawing is where the shape stops and holds while you hold the note, as on Zadar; let go and it plays on from there. Drag it in from the right edge (all the way back to the edge turns it off) and double-click it to turn it off. On ENV 1 the rest of the shape is the release: it plays out first, then the Release fade follows. The ADSR and Acoustic banks draw the classic stages, so a sustain point on their flat part makes an ADSR you can see.

Envelopes moving envelopes. Env 1, 2 and 3 are sources, and each envelope's Time, Warp, Bend, Level, Sustain Point and Chain Phase (and its ADSR stage times) are destinations, so one envelope can stretch, lean or bend another while it plays, or itself. Time is read as a rate, so a change speeds the shape up or slows it from where it is, never jumps it. Chain Phase is read when the chain hands over. *Breathing Planet* is built this way.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
Time	`1 ms – 30 min`	`2 s`	How long one pass takes, with **FREE** lit. ROUTE
Length	`1/16 – 64 bars`	`1 bar`	With **SYNC** lit, one pass in the host's tempo instead.
Warp	`-100 – +100%`	`0%`	Drag sideways: + pushes the shape later, – pulls it sooner. ROUTE
Bend	`-100 – +100%`	`0%`	Drag up or down: + lifts the shape, – sinks it. ROUTE
Repeat	`Once, 2x – 16x, Held, Forever`	`Once`	How many passes it plays; after the last it holds where it ended. **Held** repeats while the note is held and finishes its pass after; **Forever** until the note ends.
LVL	`0 – 100%`	`100%`	The shape's height, as Zadar's level: it scales the whole contour down. ROUTE
Sustain	`0 – 100%, Off`	`Off`	The dashed line: where the shape holds while the note is held. ROUTE
REV	`Off, On`	`Off`	Plays the shape backwards. The sustain point stays where it is from the start.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
Retrigger	`Smooth, Restart, Ignore`	`Smooth`	What a new note does while it plays: **Smooth** picks the shape up where it passes the level it is at now, so it never jumps; **Restart** starts it from the top, gliding there in 5 ms so it never clicks; **Ignore** lets it play on.
REL / VEL			ENV 1's **Release**: when you let go it fades over this while the shape keeps moving under the fade (with a sustain point, after the rest of the shape has played). ENV 2 and ENV 3 show their **Velocity**, and play on after you let go.

Chains. The box at the top right of the panel chains the envelopes in Shape mode, so one starts the next: **1 > 2**, **2 > 3**, **1 > 2 > 3**, or the same as a **Loop** back to the first. The first starts with the note; each next one waits until the one before has played its **phase** (the box beside its Retrigger) of its whole run: 0% starts it with the one before, 100% when that one ends. In a loop the first starts again from the last the same way, so two arches that start each other halfway cross-fade forever. In a chain, Held and Forever play once, so the chain moves on. An ADSR envelope in a chain just starts with the note.

UNDER THE SKY

MOTION



The MOTION tab holds the two LFOs that move the flock. They only move it: they aren't sources for routes. For a sound that moves with them, use LFO 1 or 2 at the same rate.

WINGBEAT One per creature (5 Hz sine by default). It bobs each creature across its path, by **PUSH**, and beats its wings on screen.

WIND One gust for the whole sky (a slow 0.2 Hz Smooth, Global by default). It pushes every creature of the part the same way, by **PUSH**, from a direction that turns slowly on its own.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
Shape	`Sine, Triangle, Saw, Square, S&H, Smooth`	`Sine (Wind: Smooth)`	S&H holds a random value per cycle; Smooth glides between random values.
Trigger	`Retrigger, Free, Global`	`Retrigger (Wind: Global)`	See below.
FREE / SYNC	`Rate or tempo`	`Free`	Sync follows the host's tempo at **DIVISION**; Free and Global also lock to its bar.
RATE	`0.02 – 50 Hz`	`5 Hz (Wind: 0.2 Hz)`	ROUTE
PUSH	`0 – 100%`	`20% (Wind: 30%)`	How hard it moves the creatures. ROUTE
PHASE	`0 – 360°`	`0°`	Where a retriggered cycle starts, or how far a synced one sits off the beat. ROUTE
DELAY	`0 ms – 20 s`	`0`	Fades the LFO in after each note starts. ROUTE
DIVISION	`1/32 – 4/1`	`1/4`	With triplets and dotted values.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
POLARITY	`Bipolar, Unipolar`	`Bipolar`	Bipolar swings both ways around the knob; unipolar only up from it.

ENVELOPES

LFOS



LFO 1 (4 Hz sine) and **LFO 2** (0.5 Hz triangle) are plain LFOs that only modulate. The **LFOS** tab draws each one, with a dot running along it at its rate. They have the same controls as the MOTION LFOs, without Push.

TRIGGER

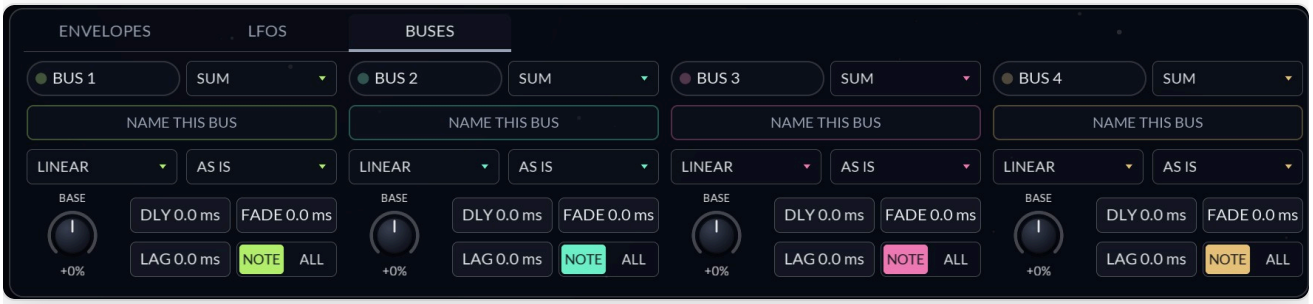
Retrigger	Each note's LFO starts at Phase. Notes are in step with how you play them.
Free	Each note's LFO picks up where the part's running LFO is, so notes are in step with each other.
Global	One LFO for the whole part. Every note hears the same value; a route can't change its rate per note.

> **ON THE GRID**
> With **SYNC** on, a **Free** or **Global** LFO is locked to your DAW's bar, not just its tempo: a 1/16 gate lands on the sixteenths wherever you start playback, and a looped bar plays the same S&H values every time round. **PHASE** shifts it against the beat. **Retrigger** starts it with each note instead, which suits notes played off the grid.

> **VIBRATO**
> Route LFO 1 to the part's **Fine** at a few percent for vibrato, and set its **DELAY** so it arrives once the note has settled. Map Press onto that route's amount and the vibrato follows your finger.

ENVELOPES

BUSES



Each part has four **mod buses**. A bus takes any number of sources, mixes them into one value, shapes it, and hands it on as a source of its own: **BUS 1** to **BUS 4**, with a chip on this tab. Build a mix once (say, Press times the mod wheel, faded in after two seconds) and route it to as many places as you like.

To feed a bus, touch a source in MAP and drag the bus's **BASE** knob, or pick the bus as a route's destination in the route list. To use it, touch its chip and drag any knob. A bus can feed another bus.

HOW IT MIXES AND SHAPES

MODE	**SUM** adds the inputs, each at its route's amount. **MULTIPLY** multiplies them, so one gates or scales another. **MIN** keeps the lowest and **MAX** the highest.
BASE	Added to the mix, −100% to +100%. With no inputs, a bus is just this value.
CURVE	**LINEAR** as it is; **EXPONENTIAL** slow near zero; **LOG** quick near zero; **S-CURVE** eased at both ends; **ABSOLUTE** folds the negative half up.
POLARITY	**AS IS**; **UNIPOLAR** turns −100..+100% into 0..100%; **BIPOLAR** turns 0..100% into −100..+100%; **INVERT** flips it.
DLY, FADE	The bus stays at zero for DLY after each note starts, then rises to full over FADE. This is how you bring in any modulation late in a held note.
LAG	Smooths the bus, so a step glides and a fast source slows down.
NOTE / ALL	**NOTE** works the bus out for each note. **ALL** makes it one value for the part, from its newest note, and times DLY and FADE from the first key down, so a note played into a held chord doesn't start the fade again.

Click the box under a bus's chip to name it. The name shows on its chip, in the route list and in MAP's hint, and is saved with the song and in your presets. An empty name goes back to "Bus 1".

> **LATE VIBRATO**
> Route LFO 1 onto Bus 1 at 100%, set the bus's DLY to 2 s and FADE to 1 s, and route Bus 1 to Fine at a few percent. Any held note now grows vibrato after two seconds. Route Press onto Bus 1 as well with MODE on MULTIPLY, and your finger sets how deep it gets.

CHAPTER 11

The Flock

What boids are, how each species flies, and how the way a creature flies becomes how it sounds.

What boids are

JSynth's creatures are **boids**, the flocking model Craig Reynolds published in 1986. No creature knows where the flock is going, and nothing steers the flock as a whole. Each creature looks only at its *neighbours*, the creatures of its own species within **RANGE**, and follows three rules:



Fig. 11.1 The three rules, for the orange creature. The dashed circle is **RANGE**, its neighbourhood; the inner pink one, 40% of Range, is too close. Each rule is a pull (pink) whose strength is its knob.

Cohesion Steer toward the middle of your neighbours. This is what holds a flock together.

Alignment Match your neighbours' heading and speed. This is what makes a flock turn as one.

Separation Steer away from any neighbour that gets too close. This is what keeps a flock from collapsing to a point.

Run those rules on every creature, 500 times a second, and shoals, swirls and murmurations come out of nothing. Nearer neighbours count more: a neighbour's pull fades smoothly to nothing at the edge of Range, and it counts for less while its own sound is fading, so a creature drifting in or out of the neighbourhood never jolts the others. JSynth adds forces of its own on top of the three rules: your note's lure, each creature's wander, the wind and the wingbeat, the walls, and a behaviour that decides how each species treats the other.

A creature's life

Each part is a **species**, and every sounding voice is one creature of it. A creature flocks only with its own kind; the other species matters only when one of them hunts or flees.

Hatching When a note starts, its creature hatches at the note's lure, pointing a random way. The sky is empty until you play.

Held While the note is held, its creature flocks with the others and is pulled toward its lure. It glows brighter the harder you press, and Press also raises its top speed.

Released When you let go, the lure lets go too. The creature flies on with the flock as its sound fades, dimming as its amp envelope releases.

Gone When the voice ends, the creature fades out and its slot is free for the next note. A stolen voice's creature goes the same way.

One step of the flock

500 times a second, each creature adds up these pulls, turns toward the sum, and moves. Its speed always stays between a fifth of its top speed and the top speed itself.

FORCE	WHAT IT DOES	SET BY
Cohesion	Toward the middle of its neighbours (away from it on Scatter).	COHESION, RANGE
Alignment	Toward its neighbours' average heading and speed.	ALIGN, RANGE
Separation	Away from any neighbour inside 40% of Range, harder the closer it is.	SEPARATE, RANGE
Lure	While the note is held, toward a point that roams around the lure on a leash: loose near it, firm when the creature strays.	LURE, WANDER, LURE X/Y
Wander	A heading of its own that drifts, so it sometimes sweeps a long curve and sometimes jinks.	WANDER
Hunt, flee	A hunter chases the nearest creature of the other species; prey shies from hunters; a fleeing species runs from the other.	Behaviour
Wind	One gust pushing every creature of the species the same way, from a direction that turns slowly.	WIND (MOTION tab)
Wingbeat	Bobs each creature side to side across its path.	WINGBEAT (MOTION tab)
Walls	Turns creatures back at the sky's edges, or carries them round.	WALLS

UNDER THE SKY

FLOCK



CONTROL	RANGE	DEFAULT	WHAT IT DOES
Behaviour	`Shoal, Scatter, Hunt, Flee`	`Shoal (B: Hunt)`	How this species treats its own kind and the other. See Behaviours below.
COHESION	`0 – 100%`	`50%`	Rule 1: how hard each creature steers toward the middle of its neighbours. ROUTE
ALIGN	`0 – 100%`	`50%`	Rule 2: how hard each creature matches its neighbours' heading and speed. ROUTE
SEPARATE	`0 – 100%`	`40%`	Rule 3: how hard each creature steers away from a neighbour inside 40% of Range. ROUTE
SPEED	`0 – 100%`	`40% (B: 55%)`	Top speed. Press raises it further, up to 80% more. ROUTE
WANDER	`0 – 100%`	`20%`	Each creature's restlessness, and how far it roams around its lure. ROUTE
LURE	`0 – 100%`	`60%`	How hard a held note pulls its creature. Low lets the flock take over. ROUTE
RANGE	`0 – 100%`	`35%`	How near counts as a neighbour: 8% to 40% of the sky's height at the default Size. ROUTE
TAIL	`0 – 100%`	`50%`	How long the ink lasts, about half a second to five seconds. ROUTE
SPREAD	`0 – 100%`	`50%`	How far the ink blooms as it ages. ROUTE
SIZE	`0 – 100%`	`50%`	How big the sky is. A bigger sky makes creatures, their Range and their speed relatively smaller, so flocks spread out. Shared.
WALLS	`Soft, Wrap, Open`	`Soft`	Soft turns creatures back near the edge; Wrap carries them round to the other side; Open lets them fly well past it first. Shared.
LURE X, LURE Y	`Pitch, Velocity, Press, Slide, Release, Random, Centre`	`Pitch, Slide`	What places a note's lure along each axis.

Behaviours

Shoal Flies with its own kind: cohesion, alignment and separation as set. The flock to start from.

Scatter Keeps apart: cohesion pushes creatures away from their neighbours' middle instead of toward it, so a chord spreads across the sky and Nearest stays low.

Hunt Chases the nearest creature of the other species it can see, up to 2.5 times its Range away. A hunter that gets within reach (about 6% of the sky's height) catches it, which turns on Bite. Every creature shies away from a hunter of the other species a little, so prey scatters as a hunter comes in.

Flee Runs hard from any creature of the other species within its Range, whatever that species is doing.

Behaviour only matters between species, so it needs both parts playing: play mode **LAYER**, or any mode that sends notes to both. A new instance starts with A on Shoal and B on Hunt.

The lure

A held note pulls its creature toward a point that roams slowly around the lure, on a leash: loose near the lure, so the flock and the wander shape the path there, and firm when the creature strays. Each creature roams on its own two slow orbits, so no two trace the same path. **LURE X** and **LURE Y** choose what places the lure: by default low notes sit left and high notes right (Pitch, over four octaves from C2, and bend moves it), and more Slide sits higher. Try Lure Y on *Press* to make pressing harder send creatures upward, or Lure X on *Random* to scatter a chord across the sky. A lure follows its note as it plays, so sliding or pressing moves it live.

How the flock reaches the sound

The flock changes nothing in the sound by itself, so a plain patch is plain however it flies. It gets in three ways, and only these:

Sources	Each creature is a modulation source for its own note: **SPEED**, **NEAREST**, **HEADING** and **TURN** (Chapter 6). Touch one under the sky and drag any knob, as with an envelope.
Between creatures	**FLOCK FM**, **TUNE PULL** and **BITE** on the FM tab let creatures hear each other (below).
X Pan	**X PAN** on the VOICE tab pans each voice by where its creature is, left to right.

So the flock's knobs shape the sound through those. This is what each one does to what you hear:

TURN UP	THE CREATURES	WHAT YOU HEAR
COHESION	Bunch together.	Nearest climbs and more creatures are in each other's range, so Flock FM rings harder and Tune Pull has more to pull toward. With X Pan, the stereo image narrows.
ALIGN	Fly as one, turning together.	Heading and Turn move together across a chord, so routes from them move every note alike. Tune Pull engages, since it only pulls creatures heading the same way.
SEPARATE	Keep their distance.	Nearest stays lower and steadier, and Flock FM gentler.
SPEED	Fly faster.	Every source moves faster: neighbours come and go sooner and X Pan sweeps quicker. The Speed source itself is measured against the top speed at full Press, so a creature you don't press stays below about 55%, and pressing fills it.
WANDER	Drift, jink and roam wider.	Turn gets busy and Heading wanders, so routes from them get livelier; the flock holds together less.
LURE	Stay where you play.	Each creature's place follows its note, so X Pan follows Lure X (by default the pitch, low left and high right). Turn it down and the flock carries creatures off.
RANGE	See farther.	The three rules, Nearest, Flock FM and Tune Pull all reach farther, so creatures hear each other from further apart.
SIZE	Spread out in a bigger sky.	Fewer neighbours, lower Nearest, slower sweeps. Shared by both species.
Behaviour	Shoal, scatter, chase or flee.	Shoal gives a high, steady Nearest; Scatter a low one and a clean sound; Hunt and Flee give sudden bursts of Speed and Turn, and Hunt drives Bite.
WALLS	Turn back at the edge, or wrap round.	With Wrap, a creature leaving one side reappears at the other, so X Pan jumps across the stereo field. Soft and Open sweep it back.

TURN UP	THE CREATURES	WHAT YOU HEAR
TAIL, SPREAD	Leave longer, wider ink.	Nothing. Ink is only drawn.

Modulating the flock

Every flock knob but Size can take routes, and a route moves each creature by its own note. A new part already routes Press onto Speed, so the note you lean on flies faster. Try Press onto **SEPARATE** and the note you press breaks formation while the rest hold together; Velocity onto **WANDER** and hard-struck notes fly wild; an envelope onto **LURE** and each creature is held at its place as the note starts, then let loose into the flock. Behaviour, Lure X and Lure Y, Size and Walls can't be routed.

The **MOTION** tab's two LFOs (Chapter 10) move only the flock. The wingbeat shows a little in Turn, and the wind in Speed and Heading, so route those to hear them.

Between creatures

Three knobs on the **FM** tab connect the flock to the sound directly. All start at 0, so the flock changes nothing until you ask it to.

FLOCK FM Each creature listens to up to three neighbours of its own species, weighted by how near they are, and their Osc 1 frequency-modulates its Osc 2. A tight flock rings; a loose one is clean. When a creature gains or loses a neighbour, the change fades in over about 10 ms, so it never clicks. For it to be heard, Osc 2 must be in the sound: give it level, or route it into Osc 1 on the FM grid. It needs at least two notes of the same part.

TUNE PULL For each neighbour heading within 30° of the same way, the creature is pulled toward the nearest just interval from that neighbour's pitch (unison, minor third, major third, fourth, fifth, major sixth, octave), up to ±30 cents at 100%, over about 150 ms. Chords played on a flock that flies together settle into just intonation; break formation and they drift back.

BITE While one of this part's hunters holds a creature of the other species, the hunter's Osc 1 ring-modulates it, by Bite times how caught it is. Set Bite on the hunting part. A catch takes hold in about 20 ms and lets go over half a second, so each bite rings out as the prey escapes.

> **HEAR THE RULES ONE AT A TIME**
> Load *Init*, set OSC 2's LEVEL to half, FLOCK FM to 60% and LURE to 20% so the flock can take over, and hold a four-note chord across two octaves. Turn COHESION to 100% and the others to 0: the creatures pile together and the chord rings. Swap to SEPARATE at 100%: they push apart and it goes clean. Then ALIGN alone: they stream off together, and with TUNE PULL up the chord slides into just intonation as they line up.

Ink

Every creature leaves ink behind it in the sky, more the brighter it is. The ink fades with **TAIL** and spreads with **SPREAD**, and creatures flying through it stir it. How it looks depends on the sky's look and palette (*Chapter 6*). It is only drawn: it doesn't change the sound.

CHAPTER 12

Touch

Five dimensions of every note, each shaped and smoothed, each a source.

UNDER THE SKY

TOUCH



TOUCH	WHERE IT COMES FROM
VELOCITY	How hard the note was struck: its velocity.
PRESS	Pressure: MPE channel pressure or the CLAP pressure expression. It also makes the creature faster and brighter.
GLIDE	The finger's bend, as −100% to +100% of the bend range.
SLIDE	Up and down the key: MPE CC 74 or the CLAP brightness expression.
RELEASE	How fast the note was released: its release velocity, 0 while the note is held. Route it to the amp envelope's release, say, so a quick lift cuts the note short.

Each has a **CURVE** (−1 to +1): positive bows it up, so it responds more early in the travel, and negative bows it down. And a **SMOOTH** time (8 ms by default, 2 ms for Glide so slides track the finger), which irons out a controller's steps.

Under them, the rest of how you play. The whole tab is shared by both parts.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
ROUND	`0 – 100%`	`0%`	Glide Rounding: once a finger stops sliding, pull its pitch toward the nearest semitone, by this much. While the finger moves, the pitch follows it exactly.
ROUND TIME	`0 ms – 20 s`	`150 ms`	How long that pull takes.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
MGR BEND	`1 – 96 st`	`2 st`	The bend range of the MPE manager channel (channel 1), which bends every note.

The touch chips

The five chips in the header light as you play, showing the newest note of the part you are editing. In MAP mode they are sources like any other: touch **PRESS** and drag a knob to make pressure move it.

> **NO MPE CONTROLLER?**
> Velocity comes from any keyboard. A plain keyboard's channel pressure is Press for every note, and its pitch wheel bends every note by MGR BEND. In the sky, a click sets Slide by how high you click and dragging bends.

CHAPTER 13

Modulation: MAP and Routes

There is no matrix page. Touch a source, drag a knob.

Mapping

01 **Touch a source**

Click any source chip: the touch chips in the header, the envelope, LFO and bus chips on their panels, or the flock's under the sky. That turns MAP on: the source chips pulse, the one you touched lights, and the sky tells you to drag a knob.

02 **Drag a knob**

Drag any knob up or down to set how much the source moves it, from −100% to +100%. A ring in the source's colour shows how far it reaches. Drag more knobs to send the same source to them too, or touch another source.

03 **Touch the lit source again**

To finish. Knobs go back to setting their own values, and the rings go away.



Fig. 13.1 MAP with a source armed: the knobs it moves show rings in its colour.

IN MAP MODE, WITH A SOURCE ARMED	ON A KNOB
Drag	Set that source's amount on this knob. A route dragged back to 0 is removed.
Double-click	Flip the route's sign.
Right-click	Remove the route. With no source armed, removes every route to that knob.

The value boxes (an envelope's TIME, REL, VEL and phase) take routes the same way: drag up or down, and a bar along the box's foot shows how far the source moves it. A Shape envelope's drawing takes a sideways drag as a route onto its Warp and an up-or-down one onto its Bend.

Only the armed source's rings show; touch another source to see its routes. Nearly every knob on a part's panels can take routes: the flock's knobs, the FM grid, the envelopes and LFOs, the filters and the oscillators. The exceptions are Transpose, the part's Bend range, Velocity, and the shared ones: Size, Width, Master and the TOUCH tab.

Sources

SOURCE	VALUE
Env 1, 2, 3	0 to 100%, each note's own.
LFO 1, 2	−100% to +100% (or 0 to 100% when Unipolar). Per note unless Global.
Velocity, Press, Slide	0 to 100%, each note's own.
Release	The note's release velocity, 0 to 100%: 0 while it is held.
Glide	−100% to +100%: the note's bend.
Key	The note: −100% four octaves below middle C, 0 at middle C, +100% four octaves above.
Speed, Nearest, Heading	The flock's own, 0 to 100% (Chapter 6).
Turn	The flock's turning, −100% (left) to +100% (right) (Chapter 6).
Random	A new value per note, −100% to +100%.
Mod Wheel	CC 1 on channel 1.
Macro 1 to 4	Four host parameters, for automation or a controller.
Bus 1 to 4	The part's mod buses, −100% to +100%, by their own names (Chapter 10).

Key, Mod Wheel and the macros have no chip on the panel; choose them in the route list. Random's chip is under the sky with the flock's.

The route list



Fig. 13.2 ROUTES: every route of the part in use, with its source, destination and amount.

ROUTES under the sky opens, over the sky, the list of the part's routes in use, sixteen to a page. Each row has the source, the destination and the amount; change any of them here. **+ ADD** starts a new route (Env 2 onto Filter 1 Cutoff, at 0) to edit, and **X** removes one. The arrows turn the pages. Each part has 64 routes; the list shows how many are used.

The list reaches what MAP can't: sources without a chip, and destinations with no knob on the panel, such as an oscillator's Phase, an envelope's Delay and Hold, and every route's own amount.

Routes onto routes

A route's amount is itself a destination. Route Press onto the amount of a route from LFO 1 to Fine, and the vibrato deepens as you press. Routes onto amounts are worked out first, so they always take effect on the same tick.

How far a route moves a knob

A route moves a parameter within its own range: +100% from a source at full moves a knob from its lowest setting to its highest, and +10% moves it a tenth of the way. Routes add together, and the result stops at the knob's ends. Ratio moves by musical intervals, so a small route onto Ratio detunes evenly at every setting.

CHAPTER 14

Factory Presets

A hundred and sixteen sounds in 12 categories. The first eleven are the kinds of sound a session reaches for, keys to effects, each voiced from scratch on JSynth's engine and played straight from a keyboard; the last, *Flock*, holds the drones, textures and landscapes the creatures play themselves.

The preset bar works like JVerb's: the arrows step through the presets, scrolling over the bar does too, and clicking the name opens the menu by category. **SAVE** saves the current sound under a name of your own. The **star** beside the name makes the current preset a favorite, and *Favorites* at the top of the menu lists every starred one, so the sounds you reach for are one click away. A factory preset sets every parameter it doesn't mention to its default, so nothing from the last sound (a route, say) carries over. The first eleven categories are sounds to play, voiced the way a big commercial bank lays them out: each works from any keyboard, velocity shapes its attack and tone, and the mod wheel does the obvious thing for the sound (vibrato on a lead, rotor speed on an organ, tremolo on a piano, brightness on a pad), with Press and Slide adding to that on an MPE controller. The flock leaves them alone. The last category, *Flock*, keeps the drones, textures and landscapes the creatures play. The sequences play in time with the host, and the drums are tuned to be played from C2 up. Every preset is level-matched, so stepping through the bank doesn't jump.

8 PRESETS

Keys

Electric pianos, a clavinet, a felt upright and a harpsichord: velocity sets how hard they bark, and the decay shortens up the keyboard.

Tine Piano	A tine electric piano: two operators for the body, a ×14 one for the tine's ping, both harder the harder you play, and a decay that shortens up the keyboard. The wheel brings in tremolo.
Suitcase	A suitcase piano: a rounder bark, a little tube on hard notes, and a stereo tremolo that pans across the field. The wheel sets how deep the tremolo pans.
Reed Piano	A reed piano: a triangle barked by a 1:1 modulator and pushed into tube drive by the strike, through a soft low pass. Bright and nasal when you dig in, mellow when you don't.
Clav Funk	A funky clavinet: a narrow pulse and a saw through a high pass and a snapping low pass, short and percussive, quiet when you let go. Velocity opens the snap; Slide moves the pickup.
Felt Upright	A felt upright: a triangle and its octave with a soft hammer of noise, a low pass that opens with the strike and closes as the note dies, and a gentle detune between the strings.
Crystal Keys	A glassy FM piano: a ×7 bell partial over the body that rings on hard notes, an octave of sine underneath, and echoes a dotted eighth behind.

Harpsichord	A harpsichord: a narrow pulse and a saw an octave up, plucked bright through a high pass, with the quick damped release of a jack falling back. It barely follows velocity, as the real one.
Lo-Fi Keys	A worn tape of an electric piano: the tines, a slow wow, hiss and a darker top, a little soft drive. The wheel deepens the wow.

5 PRESETS

Organs

Tonewheel drawbars with a key click and a rotary speaker the wheel speeds up, pipes in a stone church, a combo organ, and a harmonium Press pumps.

Jazz Drawbars	A jazz tonewheel organ: 16', 8' and 2⅔' drawbars, a key click and a decaying third-harmonic percussion, through a tube and a rotary speaker shared by every note. The wheel speeds the rotor from chorale to tremolo.
Gospel Organ	A full gospel registration: every drawbar out, driven hard, the rotor already spinning fast. The wheel slows it down; Press pushes the drive.
Cathedral Pipes	A pipe organ in a stone church: a square principal, an octave and a sub, a breath of wind, slow to speak and slower to leave the room.
Combo Organ	A 60s combo organ: a square and a bright octave saw with a quick vibrato on every note. The wheel deepens the vibrato.
Harmonium	A harmonium: two saws a musette apart and a pulse below through a reedy peak, breathing in with the bellows. Press is the bellows: lean in and it swells.

8 PRESETS

Bells and Mallets

Struck metal and wood from FM: glockenspiel, marimba, vibraphone, tubular bells, kalimba, steel pan, music box and a crystal bell.

Glockenspiel	A glockenspiel: a bright struck bar, an octave up, with its inharmonic ×2.76 overtone fading first and a ping from the mallet.
Marimba	A marimba: a warm wooden bar, its strong fourth partial knocked out by the mallet and gone in a moment, a soft woody click, and a decay that shortens up the range.
Vibraphone	A vibraphone: a long metal bar with its ×4 overtone, a soft mallet, and the motor's tremolo, which the wheel turns up.
Tubular Bells	Tubular bells: a ×3.5 modulator that rings and slowly fades out of a long carrier, with an octave partial a hair sharp. Hit it hard for the clang.
Kalimba	A kalimba: a soft thumb on a metal tine, a quick ×3 blip and a faint inharmonic overtone, with echoes behind.
Steel Pan	A steel pan: a ringing note with its octave and twelfth, a bright metallic attack that settles quickly, and a slight pitch lift as it's struck.
Music Box	A music box: a high triangle with a faint ×3 tine and a long ringing tail, a little uneven.
Crystal Bell	A crystal bell: an inharmonic ×1.41 FM ring, high glints at ×7.13, and a slow shimmer between two carriers a few cents apart, in a big room.

8 PRESETS

Plucks

Short sounds for lines and chords, from the classic filter pluck to strings plucked in a comb tuned to the note.

Classic Pluck	The classic synth pluck: two saws a few cents apart, a resonant filter that snaps shut, and echoes a dotted eighth behind. The wheel lengthens it.
Square Pluck	A hollow square pluck, round and woody, its filter snapping with a little resonance.
Nylon String	A nylon string, plucked: a short burst of noise and saw rings a comb tuned to the note, the way a string rings, then a low pass softens it. Harder notes pluck brighter.
Harp	A concert harp: the same plucked comb, softer, its burst a triangle, ringing long into a hall.
Pizzicato	String-section pizzicato: two saws a few cents apart through a quick low pass, short and bouncing.
Bell Pluck	A bell pluck: a ×2 FM ping that falls away fast into a pure tone, with a faint fourth partial.
Mallet Pluck	A mallet pluck: a triangle and its octave through a resonant filter that pings shut, like a soft synth marimba.
Chord Stab	A house chord stab: saws and a square an octave down through a resonant low pass that bites and closes, short enough to play on the offbeat.

12 PRESETS

Bass

Mono basses from a clean sub through analog, ladder, acid, FM, Reese and wobble to a fingered upright.

Sub Bass	A clean sub: a sine with a touch of triangle and soft drive, so it still speaks on small speakers. The wheel adds grit.
Analog Bass	An analog mono bass: a saw and a square an octave down into a ladder that snaps open with the strike, a little drift between them. Harder notes accent the filter.
Ladder Punch	A fat ladder bass: three saws (one an octave down) pushed into the ladder's own drive, with a punchy, short filter envelope.
Acid Line	Acid: a saw into a squelching, driven ladder that slides between held notes; harder notes accent the filter. The wheel turns up the squelch and Press opens it.
FM Bass	A DX bass: a 1:1 pair with feedback, its bite dying away fast, and a sine sub an octave down.
Reese	A Reese: two saws a dozen cents apart, beating slowly over a sine sub, held together by drive. A slow LFO breathes the detune; the wheel opens it up.
Rubber Bass	A rubbery square bass: a square and a sub through a resonant low pass that bounces shut, with a slight pitch blip.
Wobble Bass	A wobble bass: saws and a sub in a driven low pass that an LFO in time with the host opens on every eighth. The wheel deepens the wobble.
Pluck Bass	A short plucked bass for busy lines: a saw and a sub, the filter and the amp both closing fast.
Growler	A growling bass: a saw ring-barked by a 1:1 modulator that an LFO in time with the host pulls open every quarter, through tube drive and a ladder.

Finger Bass	A fingered upright: a round triangle and its octave, a thump in the filter, and a decay that lets each note ring and fall.
Octave Bass	An octave bass: a saw and a pulse an octave up, opening together, for disco and synth-pop lines.

9 PRESETS

Leads

Mono leads that glide, with a vibrato that waits a moment and deepens on the wheel: saw, square, sync, unison, chip, a whistle, a flute and one made for MPE.

Saw Lead	A singing saw lead: two saws and a quiet octave, legato with a glide, a vibrato that waits a moment, and echoes. The wheel deepens the vibrato; Press opens the filter.
Square Lead	A hollow square lead with a sub, a short glide and the same waiting vibrato.
Sync Lead	A sync lead: Osc 1 only keeps time, and the synced saw's pitch sweeps down from a scream as each note starts. The wheel and Press push the sweep higher.
Whistle	A whistle: a pure sine with a breath of air, a lazy glide and a gentle vibrato.
Unison Lead	A wide unison lead: three saws spread across the field, a touch of drive, a quick glide.
Chip Lead	A chiptune lead: a thin pulse, no filter, a fast vibrato that comes in after the note.
Screamer	A screaming lead: a saw and a square into a hard-driven, resonant ladder. Press opens it up and makes it scream; the wheel adds vibrato.
Breathy Flute	A breathy flute: a sine with a little second harmonic, breath through a band pass at the note, and a vibrato that blooms. Press blows harder.
Expressive Lead	A lead built to be played with an expressive controller: Press opens it and adds vibrato, Slide narrows the pulse, and a wide bend lets you slide between notes. From a keyboard the wheel does the vibrato.

12 PRESETS

Pads

Held chords: warm analog, string machine, glass, choir, PWM, sweeping, dark, airy, evolving and bright unison.

Warm Analog Pad	A warm analog pad: two saws and a sub pulse, a low pass that opens slowly after the chord and breathes a little, and drift between the voices.
String Machine	A string machine: three saws, an octave on top, through an ensemble of two slow and quick pitch wobbles, panned wide.
Glass Pad	A glass pad: soft FM with a ×3 modulator that slowly comes and goes, and a second carrier a few cents off for a slow beat. Press brightens it.
Choir Aah	A choir singing "ah": a saw and a pulse through two fixed formants, with a slow vibrato and a breath. The wheel leans the vowel toward "oh".
PWM Pad	A pulse-width pad: two pulses whose widths two slow LFOs sweep in opposite ways.
Sweep Pad	A sweeping pad: saws in a resonant low pass that sweeps up for four seconds after the chord and settles half way.

Dark Pad	A dark pad: saws an octave down and a sine sub in a steep, low filter, moving slowly.
Air Pad	An airy pad: a soft triangle and its octave, washed in high, filtered air.
Evolving Pad	An evolving pad: a saw and a pulse whose filter and width two long looping Shape envelopes keep moving, out of step with each other, so a held chord never repeats itself.
Bright Unison	A bright unison pad for trance and pop: three saws spread across the field, cut below and open above.
Soft Sine Pad	A soft sine pad: a sine, a quiet octave and a triangle a few cents off, slow to bloom.
Vowel Pad	An "ooh" pad whose vowel slowly opens toward "ah" and back on an LFO.

8 PRESETS

Strings and Brass

Ensembles and soloists: strings fast and slow, a cello, synth brass and stabs, soft horns and a trumpet.

Ensemble Strings	An ensemble of strings: saws in a gentle ensemble, a bowed attack that's quicker on harder notes, and a vibrato that comes in late.
Slow Strings	Slow strings: the same section further back, swelling in over a second and a half.
Solo Cello	A solo cello: one bowed voice, legato, its body a peak in the low mids, a vibrato that blooms, and the bow's hiss. Press leans on the bow.
Synth Brass	Synth brass: two saws through a low pass that blooms a beat after the note speaks, a slight scoop up into pitch, and deeper with velocity.
Brass Stab	Brass stabs: brighter and shorter, a punch of filter for hits on the beat.
Soft Horns	Soft horns: a saw and a triangle, mellow and slow to speak, like a French horn section.
Solo Trumpet	A solo trumpet: one bright legato voice that blooms with the breath, a scoop into each note and a late vibrato. Press is the breath.
Tremolo Strings	Tremolo strings: the ensemble bowed fast, a quick swell on every stroke.

8 PRESETS

Arps and Sequences

Hold a chord and it plays in time with the host: gates, pulses, octave arpeggios and sample-and-hold.

Trance Gate	A trance gate: bright unison saws chopped into sixteenths in time with the host.
Pulse Sequence	An eighth-note pulse: a plucked pulse retriggered every eighth, its filter stepping to a new place each time.
Random Filter	A sample-and-hold filter: a saw into a resonant ladder whose cutoff jumps to a new random step every sixteenth.
Octave Arp	An octave arpeggio: each held note leaps through four octaves an eighth at a time, struck as sixteenths.
Echo Pluck	A dotted delay pluck: quarter-note plucks fed into dotted-eighth echoes, so a held chord plays a rolling rhythm.

Pad Chopper	A chopped pad: a warm pad gated in a syncopated pattern of eighths, with the filter stepping.
Data Bleeps	Computer bleeps: an FM blip whose modulator jumps to a new ratio every sixteenth.
Wobble Arp	A wobbling arp: a resonant saw opened and closed eight times a bar by a synced Wobble 8.

8 PRESETS

Percussion

A kit to play from C2 up: kick, snare, hats, tom, clap, cowbell and rim shot.

Analog Kick	A kick drum: a sine whose pitch drops fast from a knock, a little drive, tuned by the key.
Snare	A snare: a pitched body and a burst of bright noise, each through its own filter.
Closed Hat	A closed hi-hat: high, thin, metallic noise, gone in a moment.
Open Hat	An open hi-hat: the same, left to ring.
Tom	A tom: a sine and a triangle that drop in pitch as they ring, tuned by the key.
Hand Clap	A hand clap: a band of noise struck several times in quick succession.
Cowbell	A cowbell: two square tones at fixed pitches through a band pass, short and bright.
Rim Shot	A rim shot: a high sine knock and a click of noise, very short.

8 PRESETS

FX

A riser and a downlifter, a laser, wind, surf, a UFO, a siren and thunder.

Riser	A riser: noise and a saw climbing through a high pass that opens over four seconds.
Downlifter	A downlifter: the same falling away, high to low.
Laser	A laser: a sine and a saw diving three octaves in a blink, into echoes.
Wind	Wind: pink noise through a resonant band pass that drifts on its own, swelling and dying.
Ocean Surf	Surf: noise rolling in and out on a slow looping Breath, its brightness rising with each wave.
UFO	A UFO: a sine whose pitch wobbles fast, ring-barked, drifting overhead.
Siren	A siren: a square swinging up and down a fifth, slowly.
Thunder	Thunder: deep, driven noise that cracks and rolls away.

22 PRESETS

Flock

The flock's own showcases, each in the room it was voiced in: drones and textures every note's creature plays, two flocks hunting each other, a bell and a theremin.

Slow Murmuration	Saws and a sub under a slow, wide flock: each creature's climb opens its own filter, its turns swing it across the stereo field and its speed brings up the sub. A 9 s Breath loop makes the flock restless and deepens the climb as it peaks.
Aurora Choir	A choir in two formants: each creature's climb opens its vowel from "oo" toward "ah", its turns brighten the second formant, and Tune Pull draws the chord into just intervals. Flying fast deepens the vibrato, and a slow Twin Hills loop lets the vowels swing wider, then settle.
Weightless	FM glass that floats: the faster each creature flies the brighter its FM, its climb swells a sub an octave down, and its turns pull its carrier a few cents either way so the chord beats. A 20 s Arch loop speeds the whole flock up and lets it rest.
Tidal Breath	Breath through a resonant band pass at the note: a 17 s Surge loop strengthens the wind, the gusts carry every creature at once, and how fast each is carried brings up its breath and rings its band; its climb swings the band up and down.
Halo	A ring of light: a triangle and air through a comb tuned to the note. Each creature's climb lengthens its ring, its turns shade the ring a few cents off the note, and how near its neighbours are thickens the air. A 25 s Shimmer loop brightens the whole halo.
Breathing Planet	Envelopes that move each other, under the flock: a 12 s Breath on Env 2 opens the filter and quickens Env 3's wobble as it fills; each creature's climb leans its breath early or late, and its speed sets how deep its wobble runs.
Cirrus	Soft sines that ring each other: Neighbour FM makes a gathered flock shimmer and a scattered one fall pure. A 22 s Arch loop pulls the creatures together and lets them drift apart, so the shimmer comes and goes; each one's climb lifts it into its octave.
Abyssal	A sub and a growl in a low ladder: each creature's climb opens the ladder, how near its neighbours are drives it, and a 23 s Surge loop sends the flock racing, so the growl rings when it does.
Obsidian Choir	A low, dark choir in two formants near "oo": each creature's climb opens its vowel toward "oh", its speed adds breath, and Tune Pull holds the chord in just intervals. A 29 s Bow loop deepens how far the vowels move.
Gravity Well	Creatures falling together: a Late Swell on Env 2 raises Cohesion and the range over 16 s, so a held chord gathers; the nearer each creature's neighbours, the more FM and resonance it gains, until the drone tightens into a ringing knot. Let go and it scatters.
Firefly Field	Half-sine sparks on a Sparkle amp, in a fast, scattered flock: the faster a creature flies the denser its sparks, its climb lifts their pitch a little, and crowding brightens them.
Rain on Glass	FM glass struck by Sparkle, each creature its own rainfall: how near its neighbours are deepens the FM, its turns pan it, its speed quickens the drops, and a 20 s Gusts loop brings squalls, flying the flock faster as it rains harder.
Embers	Pink noise in a comb tuned to the note, crackling on a Flicker amp: each creature's climb bends the comb up and down, its speed whitens the noise, and a hot creature (fast and crowded at once) rings longest.
Aeolian Harp	A wind harp: noise through a ringing comb at the note in a strong, gusting wind, and how fast each creature is carried sets how loud its string sings; its climb brightens the bow and its turns lean the string a few cents off.

Turbine	A turbine spinning up: a Slow S on Env 2 raises the flock's speed over 8 s, pushing the FM modulator's ratio up as it goes, and each creature's speed pushes it further, a whine climbing out of the hum; its turns add feedback, its climb opens the filter. Let go and it winds down.
Scrapyard	Struck scrap: inharmonic FM (×3.73 and ×7.11) on a looping Collapse as the amp, each strike lower than the last; each creature's turns knock its ratios off true so no two hits ring alike, its speed sets how fast the pile collapses and its crowding how hard it clangs.
Engine Room	Engines idling: detuned saws throbbing on LFO 1, whose rate is each creature's speed (an engine revving) and whose depth is how near its neighbours are; a 26 s Wander leans on the drive.
Glass and Iron	Press crossfades a calm sky into a storm: part A is FM glass drifting in a slow shoal, part B a driven comb of noise and saws whose hunters bite. Each part's creatures play its own sound: A's climb rings its glass, B's speed drives its grind and its turns hammer its comb.
Predator Weather	Layer: a soft saw drone in a slow shoal, and high FM bells (part B, an octave up) that hunt it. When a bell catches a creature of the drone it bites, ring-modulating it; the bells' speed brightens them and their turns shift their ratio.
Two Flocks	Layer: two flocks that ring each other. A flees and B hunts it a fifth up, both with Neighbour FM, so each species shimmers when it gathers; A's climb opens its filter, B's speed its FM.
Flock Bells	Tubular bells: a ×3.5 modulator over a long ring and a carrier an octave up; each creature's turns knock the modulator off its ratio, so a held bell's colour wavers.
Sky Theremin	A theremin: Press lifts the creature up the sky; flying upward bends it sharp, diving flat, through a slow lag, and the wingbeat, deepened by its speed, is its vibrato.

> **INIT**
> *Init* is every parameter at its default: a saw through a low pass on part A, with the three starting routes (Env 2 and Press on the cutoff, Press on the flock's speed).

Your presets

User presets are small text files ending `.jspreset`, saved in:

macOS	`~/Library/Application Support/JSynth/Presets`
Windows	`%APPDATA%\JSynth\Presets`
Linux	`~/.config/JSynth/Presets (or $XDG_CONFIG_HOME/JSynth/Presets)`

Copy them between machines freely. A preset from a newer JSynth loads in an older one with any parameters it doesn't know skipped.

Favorites are kept in `Favorites.txt`, one preset name per line, in the folder above `Presets`. Every JSynth you open shares them, and they last between sessions.

CHAPTER 15

Sound Design Recipes

Starting points that show off what the flock can do. Each starts from Init unless it says otherwise.

DX electric piano

Two sines, a 1:1 pair for the body and a ×14 for the bark.

```
Osc 1 Sine · Osc 2 Sine Ratio ×1 Level 0
Osc 3 Sine Ratio ×14 Level 0
Env 1 A 1 ms D 5 s S 0 R 400 ms
Env 2 → FM 2 > 1 +30% · Env 3 (D 60 ms) → FM 3
> 1 +35%
Velocity → FM 2 > 1 +20% · Filter 1 open
```

Or load *Tine Piano*.

A flock that rings

Neighbours modulate each other, so a held chord shimmers as it gathers and breaks.

```
Osc 2 Sine Ratio ×3.5 · FM 2 > 1 40%
FLOCK FM 70% · TUNE PULL 40% · COHESION 70%
Nearest → FM 2 > 1 +30%
X PAN 60%
```

Or load *Cirrus*.

Predator and prey

Two species; every catch rings, and the prey's filter opens as its own kind closes in.

```
Play mode LAYER
A: Flee · Nearest → F1 Cutoff +60%
B: Hunt · Speed 70% · Pulse · BITE 90% · Transp
-12
```

For hunters that ring as bells, load *Predator Weather*.

Turning pad

The sound leans as each creature banks.

```
Osc 1 Alt Sine · FB 1 25%
TAIL 100% · SPREAD 90% · LURE 35%
Turn → F1 Cutoff +50% · Turn → FM 2 > 1 +40%
```

Slow Murmuration pans on the same Turn.

Wind through a choir

A strong gust pushes the flock while LFO 2 sweeps the filter.

```
Saws, Noise 35% · F1 Band Pass, Res 40%
Wind Push 90% · Wind Rate 0.1 Hz
LFO 2 → F1 Cutoff +40% · Slide → F1 Res +30%
```

For a choir the flock sings itself, load *Aurora Choir*.

Pressure vibrato

Vibrato only where you press, arriving after the note settles.

```
LFO 1 5.5 Hz, DELAY 350 ms
Route LFO 1 → Fine at 0
Press → that route's amount +6%
```

Sky Theremin takes its vibrato from the wingbeat instead.

A stereo murmuration

The flock spreads a pad across the speakers as it turns.

```
Two saws, detuned · ALIGN 80% · SPEED 25%
X PAN 80% · DRIFT 15%
Heading → Pan +60%
```

Or load *Slow Murmuration*.

Struck drum

A pitch drop that goes further the harder you hit.

```
Osc 1 Sine · FB 1 15% · Env 1 D 300 ms S 0
Env 3 (D 45 ms) → Osc 1 Ratio +10%
Velocity → that route's amount +10%
```

APPENDIX A

Parameter Reference

Every parameter JSynth exposes to the host. Part parameters exist once for each part, named "A ..." and "B ..." in the host.

GLOBAL

PARAMETER	RANGE	DEFAULT
Play Mode	Single, Layer, Split, MPE Zones, Velocity, Expression	Single
Split Key	C-1 – G9	C4
Velocity Split	1 – 127	80
Crossfade Source	Slide, Press	Slide
Quality	1x, 2x, 4x	2x
Master Level	-60 – +6 dB	-6.0 dB
Output Width	0 – 200%	100%
Manager Bend Range	1 – 96 st	2 st
Glide Rounding	0 – 100%	0%
Rounding Time	0 ms – 20 s	150 ms
Macro 1 – 4	0 – 100%	0%
Velocity ... Release Curve	-1 – +1	+0.00
Velocity ... Release Smoothing	0 ms – 20 s	8.0 ms (Glide 2.0 ms)
Size	0 – 100%	50%
Walls	Soft, Wrap, Open	Soft

VOICE (PER PART)

PARAMETER	RANGE	DEFAULT	
Level	-60 – +6 dB	0.0 dB	ROUTE
Pan	-100% – +100%	+0%	ROUTE
X Pan	0 – 100%	0%	ROUTE
Transpose	-48 – +48 st	+0 st (B: -12 st)	
Fine	-100 – +100 ct	+0 ct	ROUTE
Key Low, Key High	C-1 – G9	C-1, G9	

PARAMETER	RANGE	DEFAULT	
Polyphony	1 – 16	8	
Voice Mode	Poly, Mono, Legato	Poly	
Glide	0 ms – 20 s	0.0 ms	ROUTE
Glide Mode	Always, Legato	Always	
MPE Bend Range	1 – 96 st	48 st	
Drift	0 – 100%	0%	ROUTE
Velocity	0 – 100%	100%	
Output	Main, Own	Main	

OSCILLATORS (PER PART: OSC 1, 2, 3)

PARAMETER	RANGE	DEFAULT	
Wave	Sine, Triangle, Saw, Square, Pulse, Half Sine, Abs Sine, Quarter, Alt Sine, Alt Half, Camel, Alt Camel, White, Pink, Random	Saw (Osc 2, 3: Sine)	
Shape	0 – 100%	50%	ROUTE
Pitch Mode	Note, Ratio, Fixed	Note (Osc 2, 3: Ratio)	
Coarse	−48 – +48 st	+0 st	ROUTE
Fine	−100 – +100 ct	+0 ct	ROUTE
Ratio	×0.25 – ×16	×1 (Osc 3: ×3)	ROUTE
Fixed	1 Hz – 10 kHz	440 Hz	ROUTE
Sync	Off, On	Off	
Retrigger	Off, On	On	
Phase	0 – 360°	0°	ROUTE
Bend	0 – 100%	100%	
Key Track	Off, On	On	
Filter	1+2, F1, F2	1+2	
Level (mixer)	0 – 100%	100% (Osc 2, 3: 0%)	ROUTE
Pan (mixer)	−100% – +100%	+0%	ROUTE

FM, NOISE AND DRIVE (PER PART)

PARAMETER	RANGE	DEFAULT	
FM Feedback 1 – 3, FM 1 > 2 … 3 > 2	0 – 100%	0%	ROUTE
FM Type	Phase, Exponential	Phase	
Noise Level	0 – 100%	0%	ROUTE

PARAMETER	RANGE	DEFAULT	
Noise Color	-100% – +100%	+0%	ROUTE
Noise Filter	1+2, F1, F2	1+2	
Drive Type	Soft, Hard, Tube	Soft	
Drive	0 – 100%	0%	ROUTE

FILTERS (PER PART: FILTER 1, 2)

PARAMETER	RANGE	DEFAULT	
Filter Routing	Serial, Parallel	Serial	
Filter Balance	-100% – +100%	+0%	ROUTE
Type	Low Pass, High Pass, Band Pass, Notch, Peak, Ladder, Comb +, Comb –	Low Pass (F2: Ladder)	
Slope	12, 24, 36, 48 dB	24 dB	
Cutoff	20 Hz – 20 kHz	1.26 kHz (F2: 20.00 kHz)	ROUTE
Resonance	0 – 100%	20% (F2: 0%)	ROUTE
Drive	0 – 100%	0%	ROUTE
Key Track	0 – 100%	0%	ROUTE

ENVELOPES (PER PART: ENV 1, 2, 3)

PARAMETER	RANGE	DEFAULT	
Delay	0 ms – 20 s	0.0 ms	ROUTE
Attack	0 ms – 20 s	5.0 ms	ROUTE
Hold	0 ms – 20 s	0.0 ms	ROUTE
Decay	0 ms – 20 s	400 ms (Env 2, 3: 300 ms)	ROUTE
Sustain	0 – 100%	70% (Env 2, 3: 30%)	ROUTE
Release	0 ms – 20 s	300 ms	ROUTE
Attack, Decay, Release Curve	-1 – +1	+0.00	
Loop	Off, Held, Free	Off	
Velocity	0 – 100%	0%	
Time Range	Short, Normal, Long	Normal	
Mode	ADSR, Shape	ADSR	
Shape	60 shapes in 10 banks	Swell: Arch	
Shape Time	1 ms – 30 min	2.00 s	ROUTE
Shape Sync	Off, On	Off	

PARAMETER	RANGE	DEFAULT	
Shape Length	1/16 – 64 bars	1 Bar	
Warp	-100% – +100%	+0%	ROUTE
Bend	-100% – +100%	+0%	ROUTE
Repeat	Once, 2x – 16x, Held, Forever	Once	
Reverse	Off, On	Off	
Retrigger	Smooth, Restart, Ignore	Smooth	
Chain Phase	0 – 100%	100%	ROUTE
Sustain Point	0 – 100%, Off	Off	ROUTE
Shape Level	0 – 100%	100%	ROUTE
Env Chain (per part)	Off, 1 > 2, 2 > 3, 1 > 2 > 3, and each as a loop	Off	

LFOS (PER PART: WINGBEAT, WIND, LFO 1, LFO 2)

PARAMETER	RANGE	DEFAULT	
Shape	Sine, Triangle, Saw, Square, S&H, Smooth	Sine (Wind: Smooth, LFO 2: Triangle)	
Rate	0.02 – 50 Hz	5.00, 0.20, 4.00, 0.50 Hz	ROUTE
Sync	Off, On	Off	
Division	1/32 – 4/1	1/4	
Trigger	Retrigger, Free, Global	Retrigger (Wind: Global)	
Phase	0 – 360°	0°	ROUTE
Delay	0 ms – 20 s	0.0 ms	ROUTE
Polarity	Bipolar, Unipolar	Bipolar	
Push (Wingbeat, Wind)	0 – 100%	20%, 30%	ROUTE

BUSES (PER PART: BUS 1 – 4)

PARAMETER	RANGE	DEFAULT	
Base	-100% – +100%	+0%	ROUTE
Mode	Sum, Multiply, Min, Max	Sum	
Curve	Linear, Exponential, Log, S-Curve, Absolute	Linear	
Polarity	As Is, Unipolar, Bipolar, Invert	As Is	
Delay	0 ms – 20 s	0.0 ms	
Fade	0 ms – 20 s	0.0 ms	
Lag	0 ms – 20 s	0.0 ms	
Global	Off, On	Off	

FLOCK (PER PART)

PARAMETER	RANGE	DEFAULT	
Behaviour	Shoal, Scatter, Hunt, Flee	Shoal (B: Hunt)	
Cohesion	0 – 100%	50%	ROUTE
Alignment	0 – 100%	50%	ROUTE
Separation	0 – 100%	40%	ROUTE
Speed	0 – 100%	40% (B: 55%)	ROUTE
Wander	0 – 100%	20%	ROUTE
Lure	0 – 100%	60%	ROUTE
Range	0 – 100%	35%	ROUTE
Neighbour FM	0 – 100%	0%	ROUTE
Tune Pull	0 – 100%	0%	ROUTE
Bite	0 – 100%	0%	ROUTE
Tail Length	0 – 100%	50%	ROUTE
Tail Spread	0 – 100%	50%	ROUTE
Lure X, Lure Y	Pitch, Velocity, Press, Slide, Release, Random, Centre	Pitch, Slide	

ROUTES (PER PART: ROUTE 1 – 64)

PARAMETER	RANGE	DEFAULT	
Source	Off and 25 sources	Off (1–3: Env 2, Press, Press)	
Destination	181 destinations	Filter 1 Cutoff (2: Speed)	
Amount	−100% – +100%	+0% (1–3: +30, +40, +20%)	ROUTE

APPENDIX B

Specifications

Format	CLAP 1.2 instrument. Features: instrument, synthesizer, stereo.
Ports	One note input (CLAP, MIDI and MIDI MPE; CLAP preferred). Three stereo outputs: Main, Part A, Part B.
Platforms	macOS universal (10.15+ Intel, 11+ Apple Silicon), Windows (64-bit), Linux (64-bit).
Latency	None.
Parts	Two, each a complete patch and a species in the flock.
Voices	Up to 16 per part (40 slots in all, so stolen notes fade out in 3 ms). Mono and Legato with a note stack.
Oscillators	3 per voice, 15 waves (Random among them), double-precision phase. Saw, square and pulse band-limited by polyBLEP.
FM	3 × 3 grid including feedback; phase or exponential. Plus Neighbour FM from up to three neighbours.
Filters	2 per voice: TPT state variable (12–48 dB), notch, peak, 4-pole ZDF ladder, comb ±. Serial or parallel, each oscillator and the noise picking its filter.
Modulation	3 envelopes, each a DAHDSR with curves and loop or a Shape (60 contours, 1 ms to 30 min, chainable); 4 LFOs (2 move the flock); 4 mod buses; 5 touch dimensions, 4 flock sources, key, random, mod wheel, 4 macros. 64 routes per part.
Oversampling	1×, 2× (default) or 4×, decimated by polyphase IIR half-bands.
Control rate	Every 32 samples, ramped per sample. The flock steps at 500 Hz. Renders are identical at any buffer size.
Parameters	All automatable except the route sources, destinations and Quality. Continuous ones accept CLAP modulation; route destinations accept it per note (note id, key, channel or port).
State	Saved with the host project, including the preset name. States from older and newer builds load, with unknown parameters skipped and missing ones at their defaults.
Panel	1648 × 940 at 100%; 75%, 100%, 125%, 150% and 200%.
Threading	The audio thread never allocates, locks or waits on the panel after activation.

Troubleshooting

SYMPTOM	WHAT TO CHECK
JSynth doesn't appear in my host (macOS)	The download is probably quarantined. See the warning in Chapter 2, then rescan plug-ins.

SYMPTOM	WHAT TO CHECK
Only part A plays	The play mode is **SINGLE**. Choose **LAYER** or another mode in the header. Check each part's Key Low and Key High too.
Turning up the flock knobs changes nothing in the sound	By design: the flock only reaches the sound through routes and the three BETWEEN CREATURES knobs. Map a flock source, or raise **FLOCK FM**.
FLOCK FM makes no difference	It modulates Osc 2. Give Osc 2 some **LEVEL**, or route it into Osc 1 with **2 > 1**. Creatures also have to be near each other: raise **RANGE** or **COHESION**, and play more than one note.
BITE does nothing	Bite is set on the hunting part, and needs the hunter's Osc 1 and a prey of the other species: play mode LAYER (or any mode that plays both), one part on **Hunt**.
Notes are out of tune	Check **TUNE PULL** (just intervals are up to 30 cents from equal temperament), **DRIFT**, and a route onto Fine or Ratio. An MPE controller sends bend per note at the part's **BEND** range, which must match the controller's (48 semitones is the MPE standard).
Pitch bend is far too wide on a plain keyboard	A plain keyboard is on channel 1, which bends by **MGR BEND** (2 semitones). If its notes come on other channels, they are taken as MPE member channels and bend by **BEND**.
Notes cut off when I play a lot	The part is at its polyphony and the oldest notes are being taken. Raise the voices count on the VOICE tab (up to 16).
My user presets are missing	Check the preset folder for your OS (Chapter 14). Files must end in `.jspreset`.

Glossary

Creature A note, as it flies in the sky. Each sounding voice has one.

Species A part. Each part's creatures flock with their own kind.

Lure Where a held note pulls its creature.

Boids The classic flocking model (Craig Reynolds, 1986): each creature steers only by its neighbours, by cohesion, alignment and separation. See Chapter 11.

Neighbour A creature of the same species within RANGE. Only neighbours flock together and hear each other.

Hunter, prey A creature of a part on Hunt, and the creature of the other species it chases. A catch turns on Bite.

MPE MIDI Polyphonic Expression: each note on its own MIDI channel, so each can bend and be pressed on its own.

FM Frequency modulation: one oscillator wobbling another's phase or pitch at audio rate, for bright, bell-like and metallic tones.

Ratio An oscillator's frequency as a multiple of the note's.

Just interval An interval from a simple whole-number ratio, such as 3:2 for a fifth. They beat less than equal-tempered ones.

Route A source, a destination and an amount.

Ring modulation Multiplying two signals, for clangorous sum-and-difference tones.



JSynth

THE FLOCKING SYNTH · V0.15.0

github.com/jeffmhopkins/Jclap